Lab 7: Fuzzing for Input Validation Bugs

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Aim

To use property-based fuzz testing with the Hypothesis library to find input validation bugs in `self_storage.py`, and then fix them. The module has three functions: `parse_rental`, `duration_average` and `payable_rent`. (In this repository the application file is `self_storage.py` and the test file is `test_self_storage.py`.)

Environment: Python 3.13.0, Hypothesis 6.170.15, pytest 9.1.1 (Windows).

Step 2: Output of the original (buggy) application

```
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7> python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): medium:6
Unit and Months: ('Medium', 6.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): 3,6,12
Average Duration: 7

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 2400|200
Payable Rent: 2632.0
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7>
```

The original code works for valid input such as `medium:6`, `3,6,12` and `2400|200`, but it does no input validation.

Step 4: Hypothesis tests run on the buggy code

```
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7>  python -m pytest test_self_storage.py --tb=no
============================= test session starts =============================
platform win32 -- Python 3.13.0, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7
plugins: hypothesis-6.170.15, cov-7.1.0
collected 3 items

test_self_storage.py FFF                                                 [100%]

=========================== short test summary info ===========================
FAILED test_self_storage.py::test_parse_rental_no_crash - AssertionError: parse_rental crashed with AttributeError: 'NoneType' object has no attribute 'split'
FAILED test_self_storage.py::test_duration_average_no_crash - AssertionError: duration_average crashed with AttributeError: 'NoneType' object has no attribute 'split'
FAILED test_self_storage.py::test_payable_rent_no_crash - AssertionError: payable_rent crashed with AttributeError: 'NoneType' object has no attribute 'split'
============================== 3 failed in 0.50s ==============================
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7>
```

All 3 tests fail. Each test fuzzes one function with random text and `None` and fails if the function raises any exception.

How the code was fixed

Each function now checks that its input is a string, checks the expected format (exactly one `:` or `|`), and wraps the numeric conversions in `try/except`. Invalid input returns `None`. The functions also reject `nan`/`inf` values and negative numbers, and `parse_rental` and `duration_average` reject zero or negative months.

Bugs found by Hypothesis

Failing inputs reported by Hypothesis (*Falsifying example*) in the full test output:

Function	Failing input	Error	Fix applied
`parse_rental`	`None`	AttributeError	Check that input is a string
`parse_rental`	`''`	ValueError (unpack)	Require exactly one ':'
`parse_rental`	`':'`	ValueError (float)	try/except around float()
`duration_average`	`None`	AttributeError	Check that input is a string
`duration_average`	`''`	ValueError (int)	try/except around int()
`payable_rent`	`None`	AttributeError	Check that input is a string
`payable_rent`	`''`	ValueError (unpack)	Require exactly one '|'
`payable_rent`	`'|'`	ValueError (float)	try/except around float()

Step 5: Output of the fixed application

```
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7> python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): medium:6
Unit and Months: ('Medium', 6.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): 3,6,12
Average Duration: 7

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 2400|200
Payable Rent: 2632.0
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7> python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): medium
Unit and Months: None

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): abc
Average Duration: None

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 2400
Payable Rent: None
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7>
```

The first run uses valid input and gives the same results as before. The second run uses invalid input (`medium`, `abc`, `2400`) and each function now returns `None` instead of crashing.

Step 6: Hypothesis tests run on the fixed code

```
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7> python -m pytest test_self_storage.py --tb=no
========================================= test session starts =========================================
platform win32 -- Python 3.13.0, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7
plugins: hypothesis-6.170.15, cov-7.1.0
collected 3 items

test_self_storage.py ...                                                                    [100%]

========================================== 3 passed in 0.87s ==========================================
PS C:\Users\Lenovo\L6_L7_51_self_storage\Lab 7>
```

All 3 tests pass against the fixed `self_storage.py`.

Source Code 1: self_storage.py (fixed application)

```
import math


def parse_rental(text):
    if not isinstance(text, str):
        return None
    parts = text.split(':')
    if len(parts) != 2:
        return None
    unit, months = parts
    unit = unit.strip()
    if not unit:
        return None
    try:
        months = float(months)
    except ValueError:
        return None
    if not math.isfinite(months) or months <= 0:
        return None
    return unit.title(), months


def duration_average(months_csv):
    if not isinstance(months_csv, str):
        return None
    try:
        values = [int(r) for r in months_csv.split(',')]
    except ValueError:
        return None
    if any(v <= 0 for v in values):
        return None
    return sum(values) // len(values)


def payable_rent(text):
    if not isinstance(text, str):
        return None
    parts = text.split('|')
    if len(parts) != 2:
        return None
    try:
        rent = float(parts[0])
        discount = float(parts[1])
    except ValueError:
        return None
    if not (math.isfinite(rent) and math.isfinite(discount)):
        return None
    if rent < 0 or discount < 0:
        return None
    total = rent * 1.18 - discount
    if not math.isfinite(total):
        return None
    return max(total, 0)


def main():
    print("==== Running parse_rental ====")
    user_input = input("Enter unit:months (e.g. medium:6): ")
    result = parse_rental(user_input)
    print("Unit and Months:", result)

    print("\n==== Running duration_average ====")
    user_input = input("Enter rental months as CSV (e.g. 3,6,12): ")
    result = duration_average(user_input)
    print("Average Duration:", result)

    print("\n==== Running payable_rent ====")
    user_input = input("Enter rent|discount (e.g. 2400|200): ")
    result = payable_rent(user_input)
    print("Payable Rent:", result)

if __name__ == "__main__":
    main()
```

Source Code 2: test_self_storage.py (Hypothesis fuzz tests)

```
from hypothesis import given, settings, strategies as st
from self_storage import parse_rental, duration_average, payable_rent


@settings(report_multiple_bugs=False)
@given(st.text() | st.none())
def test_parse_rental_no_crash(s):
    try:
        parse_rental(s)
    except Exception as e:
        assert False, f"parse_rental crashed with {type(e).__name__}: {e}"


@settings(report_multiple_bugs=False)
@given(st.text() | st.none())
def test_duration_average_no_crash(s):
    try:
        duration_average(s)
    except Exception as e:
        assert False, f"duration_average crashed with {type(e).__name__}: {e}"


@settings(report_multiple_bugs=False)
@given(st.text() | st.none())
def test_payable_rent_no_crash(s):
    try:
        payable_rent(s)
    except Exception as e:
        assert False, f"payable_rent crashed with {type(e).__name__}: {e}"
```